



Converting fractions to recurring decimals

1 Terminating decimals have a finite number of digits after the decimal point. Fill in the blank

2 Without using a calculator, work out the decimal value of $\frac{13}{20}$. 0.65 Fill in the blank

3 Select all the fractions that can be written as terminating decimals. Tick 3 correct answers

☐ $\frac{11}{15}$

☒ $\frac{11}{4}$

☒ $\frac{11}{8}$

☒ $\frac{11}{50}$

☐ $\frac{11}{30}$

4 Aisha uses her calculator to convert 0.668 to a fraction. She gets an answer of $\frac{167}{\square}$. What number should go in the square? 250 Fill in the blank



- 5 Match each fraction to its terminating decimal. You can use a calculator for this question. Write the correct letter in each box

a	$\frac{73}{160}$
b	$\frac{57}{125}$
c	$\frac{63}{140}$
d	$\frac{2281}{5000}$
e	$\frac{34}{85}$

<i>b</i>	0.456
<i>a</i>	0.45625
<i>d</i>	0.4562
<i>e</i>	0.4
<i>c</i>	0.45

- 6 Sam says that $\frac{231}{330}$ is not equivalent to a terminating decimal. Without using a calculator, explain whether Sam is correct or not. Tick **1** correct answer

- ☐ Sam is correct as 330 has factors of 3 and 11
- ☐ Sam is correct as only fractions with a denominator of 10 terminate
- ☐ Sam is wrong as 330 is a multiple of 10
- ☒ Sam is wrong as the fraction simplifies to $\frac{7}{10}$ which terminates

